Saturday afternoon direction (12:44 EDT, ~4h15m to 5pm). Casey directives: (1) DEPRECATE SP-30 outreach (no response); (2) Elie continues C4 Toeplitz closure; (3) Lyra mechanism-derivation push; (4) team continues all in-flight items; (5) WRITE PAPERS for what we know; (6) GENERATE TASKS to close all gaps + assign to team. Keeper's comprehensive gap-closure + paper-writing task structure with per-CI assignments through 5pm + multi-week roadmap.

Keeper

2026-05-30 Saturday 12:44 EDT

Saturday afternoon — gaps + papers + tasks

0. Casey directives (12:44 EDT)

1. SP-30 outreach DEPRECATED. No response from prior contacts. Drop outreach work; no further outreach drafting; Bell-Vienna stays sent but no follow-up. Substrate engineering pivots to internal work.
2. Elie: continue C4 Toeplitz commutator and FINISH (multi-week target).
3. Lyra: mechanism-derivation push (cross-pattern N_max anchoring + bulk-color Toeplitz + L4 kernel-integrals).
4. All other in-flight items continue — no work paused.
5. WRITE PAPERS for what we know — papers cover current structural knowledge.
6. GENERATE TASKS to close all gaps + assign to team — this directive.

1. SP-30 outreach deprecation (administrative)

- Task #330 (eigentone, Cs-137, W-32 outreach drafts) → DEPRECATED
- `Keeper_SP30_Outreach_Drafts_v0_1.md` retained in repo as artifact but flagged DEPRECATED in next ledger update
- Bell-Vienna (Task #329 SENT 2026-05-19) stays in sent state; no further follow-up unless response arrives
- SP-30 substrate engineering work continues at internal level (substrate cartography, experimental designs internal-use); only outreach is deprecated
- Substrate Engineering Reference Manual v0.1 stays valid as internal foundational reference

2. Comprehensive gap-closure task list (assigned per CI)

ELIE — multi-week computational closure work

Task	Description	Status	Horizon
E-C4	Explicit Toeplitz commutator [T_a, T_a^†] = α(H_?) on $H^2(S)$ for D_IV^5 — substrate-natural Cartan element identification	IN FLIGHT (Toys 3642 + 3646 filed today)	weeks-multi-week (continue + finish per Casey)
E-C5	[T_a, T_b] = N_{$\alpha\beta$} T_{α+β} root-sum structure; hermiticity/semisimplicity/tracelessness verification → su(3) vs u(3) discrimination	NEXT after C4	weeks
E-C6	Identification of SU(3) Cartan eigenvalues ↔ quark color quantum numbers	After C5	weeks
E-L4	Bulk K-type radial tower kernel-integral matrix elements for explicit T190 + T2003 + T187 + m_π + m_K derivations	OPEN	weeks (parallel to C4)
E-E11	SO(5) shell-closure with spin-orbit κ_ls = C_2/n_C = 6/5 corrections; close 6 remaining magic numbers (8, 28, 82, 126, 184)	Toy 3639 framework filed today	weeks
E-B6	K-theory of D_IV^5 explicit computation; equivariant K_K(D_IV^5) for K=SO(5)×SO(2)	mapped, computation OPEN	weeks
E-B7	Higher Wallach baryon spectrum k=7,8,9,…; full hadron resonance spectrum (Δ, N*, Λ, Σ, Ξ)	conceptual, computation OPEN	weeks
E-Engine v0.3 doc	File the actual engine v0.3 consolidation doc with §6 (Drinfeld/CPT) + §7 (bulk-color algebraic) + §8 (V-compound coproduct layer)	audit-support v0.2/v0.3/v0.4 filed; actual engine doc pending	days
E-K52a S7+	BCS Bogoliubov continuation (in_progress per task list)	in_progress	multi-month
E-Two-Tier	Two-Tier substrate-precision hypothesis (Toy 3648 today) → paper-grade development; mechanism for why TIER 2 floor at ~10^{-4}	filed today, framework	days

LYRA — mechanism derivation + theoretical closures

Task	Description	Status	Horizon
L-MechCrossPattern	Mechanism for the cross-pattern: why EW-scale observables ≈ substrate-integer / N_max? Cross-link to PMNS $\sin^2\theta$_12 (42/137) ↔ Ω_m candidate (42/137) — both via rank·N_c·g = 42	IN FLIGHT per status; continue per Casey	multi-week
L-BulkColor v0.7	Absorb K3 conditions C1 (Toeplitz-source citation Upmeier/Vasilevski with theorem/page) + C2 (AdS/CFT analogy demotion to evocative parallel)	OPEN ~30 min	hours
L-StrongUniq v1.4 file	Write up v1.4 reframe absorbing pushback: C20-C24 → CANDIDATE; C22 → DERIVED CONSEQUENCE; single-residual null-model per v1.1 §6.1; "200× stronger" retired	absorption verbal; doc file pending	hours
L-StrongUniq C15-C17	Ratify or retire candidate criteria C15-C17 (cyclotomic 6-instance, 9-element arithmetic, sister principles)	multi-week	weeks
L-StrongUniq v2.0 FORMAL	Final FORMAL Strong-Uniqueness paper — target Casey "completion" milestone	multi-week	weeks-month
L-QuasiEig v0.4	V-compound coproduct burden marker resolved when engine v0.3 V-compound layer lands (Elie's E-Engine task); v0.4 absorbs	gated on Elie	days post-Elie
L-T1 v0.3	T1 falsifier sharpening per Cal cold-read (4th-gen + extra-K-type triggers)	pending Cal	days

Task	Description	Status	Horizon
L-Vol16 §3 v0.2	Color 8-vs-3 → 8 = 3 T_a + 3 T_a^† + 2 Cartan absorption per v0.6 correction; generator-count 15-vs-26 clarification; R-matrix↔Toeplitz distinction	mechanical ~30 min	hours
L-Vol16 §4 v0.2	R-matrix-induced Toeplitz commutators → Hardy-projection-induced (distinct mechanisms)	mechanical ~10 min	hours
L-Vol16 OUTLINE v0.2	§1.5 three-genera framing fix (mechanical edit per my routing note)	mechanical ~5 min	hours
L-Spectral v0.3	Absorb v0.6 Cartan-Weyl gluon-octet correction (3 T_a + 3 T_a^† + 2 K-Cartan)	mechanical ~15 min	hours
L-Generation #414 mechanism	Generation-identification mechanism beyond count-forced (h^∨=N_c=3); spinor^3 channel structural specification	multi-week	weeks
L-L5 absolute scale	L5 framework v0.1 (4 candidates honest-negatived today; genuinely OPEN; needs new approach)	OPEN	multi-week
L-A3 Higgs/EWSB mechanism	Higgs mechanism substrate-derivation (SP-31-25)	OPEN	multi-week
L-Baryogenesis α^4	α^4 in $\eta = 2\alpha^4/(3\pi)$ sized via CP-violating amplitude minimum order	OPEN	multi-week

GRACE — catalog absorption + Mendeleev + null-model

Task	Description	Status	Horizon
G-StrongUniq v1.2 backbone	When Lyra v1.4 files, support with explicit independent-vs-derived taxonomy in catalog	pending Lyra	hours post-Lyra
G-PeriodicTable v0.9	Absorb Quasi-Eigentone v0.3 + bulk-color v0.6 Cartan-Weyl + Two-Tier precision framework	active	hours-days

Task	Description	Status	Horizon
G-MasterLedger v0.8	Saturday afternoon batch absorption	active	hours
G-Two-Route v0.2	N1 null-model toy + N2 cluster wording + N3 relation-reduction (per Cal #171)	OPEN	days
G-FalsifierLiveColumn	Live-tests column F1-F5 / T1-T5 tracking	filed v0.1, update as Elie/Lyra deliver	continuous
G-G13 v0.7	8-gluon scaffold absorb v0.6 Cartan-Weyl (3 T_a + 3 T_a^† + 2 K-Cartan)	mechanical	hours
G-MendeleevQuark	Quark-mass-sector Mendeleev scan; absorb Lyra's HONEST NEGATIVE on lepton-precision; document scheme-dependence	OPEN	days
G-MendeleevCKM	CKM Mendeleev: Cabibbo derived;	V_ud	derived-consequence;
G-NuclearCorpus	Integrate orphaned nuclear corpus into Hall framework (L4 task #404)	OPEN	days-weeks
G-17Anchor	17-anchor catalog sweep (Task #122) — every item with value 17 = N_c^3 − rank·n_C	OPEN	days
G-42Anchor	42-anchor sweep (Task #155) — every item with value 42 or function-of-42	OPEN	days
G-WallachKType	Wallach K-type position matches at dims 91, 140 (Task #157)	OPEN	days
G-CSE-RLGC	CSE-RLGC tracker Bridge updates	active	continuous
G-CrossClassMatrix	Cross-Classification Matrix 256-cell completion (~164/256 populated)	active	multi-week

CAL — methodology + cold-reads + ratifications

Task	Description	Status	Horizon
Cal-Calibration#35	Cross-CI cold-read PASS on Calibration #35 candidate; ratification to STANDING per audit-chain governance	candidate filed; cross-CI PASS needed	days
Cal-StrongUniq v1.4	Cold-read v1.4 once Lyra files; absorption-PASS or further conditions	pending Lyra	days

Task	Description	Status	Horizon
Cal-Two-Tier	Cold-read Elie Toy 3648 Two-Tier substrate-precision hypothesis; structural-vs-mechanism brake	pending	days
Cal-DepthShiftFindings	Cold-read Lyra's cross-pattern + cosmological depth as they land	continuous	continuous
Cal-#218 EXACT-vs-Mechanism	BST_Methodology_EXACT_vs_Mechanism_Distinction.md (Cal own-cadence)	OPEN	multi-week
Cal-#271 K71 EXEMPLAR	K71 EXEMPLAR audit pattern inclusion in BST_Referee_Methodology.md	OPEN	days
Cal-#272 Mode 6 threshold	Mode 6 threshold formalization	OPEN	days
Cal-#264 External material audit	Substrate-closure compliance audit	OPEN	own-cadence

KEEPER — paper coordination + ledger maintenance + audit-chain

Task	Description	Status	Horizon
K-HSLedger v0.6	Saturday EOD absorption: Calibration #35 ratification (when PASS) + outreach deprecation + paper-writing initiation	scheduled	hours
K-StrongUniq v1.4 cold-read	Cold-read PASS on Lyra v1.4 when filed	pending Lyra	hours
K-Two-Tier tier-gate	Tier-gate Elie Toy 3648 Two-Tier hypothesis	pending review	hours
K-Calibration#35 cross-CI PASS	Coordinate Cal + Lyra + Elie + Grace cold-read PASS for ratification	active	hours-days
K-CLAUDE.md refresh	Headline status block update from May 22 stale to Saturday 2026-05-30 current state	OPEN	hours
K-InvestigationBoard v0.3	Refresh from earlier (Casey priority Task #383 continues)	OPEN	hours
K-EngineV0.3 audit	K-audit Elie's engine v0.3 doc when filed; K1·2 + K1·4 anchors reserved	pending Elie	hours-days
K-PaperCoordination	Coordinate paper-writing assignments across CIs (see Section 3 below)	NEW directive	active afternoon

Task	Description	Status	Horizon
K-C12-C14 META audit	Joint Lyra+Keeper; Cal #156 Flags A+B framing refinements	OPEN	days

3. Paper-writing assignments — write what we know

Eight papers worth writing now for current substantive structural knowledge:

Paper P1 — Substrate Hall Algebra (PRIMARY)

- Subject: D_IV^5 + 5 primaries → U_q$^+$(B_2) at q=2 = GF(2) substrate Hall algebra; both Serre relations covered ([2]_2=3=N_c; [3]_4=21=N_c·g); affine $B_2^{(1)}$ pin
- Lead: Lyra (already at v0.4 PASS, submission-grade pending Cal quick-verify + proof-stage genus citation)
- Status: HIGHEST PRIORITY — most-mature; can submit within days post-Cal verify
- Target: Journal-grade

Paper P2 — 4-Piece Hopf Engine of the Standard Model

- Subject: Hopf algebra structure of substrate Hall algebra IS the SM's dynamics engine: multiplication=fusion, Green coproduct=decay, R-matrix=scattering, F-part=antimatter; β-decay E3 as worked example generalized to all SM decays
- Lead: Elie (engine v0.3 doc) + Lyra (theoretical synthesis)
- Status: Engine v0.3 doc pending Elie file; paper-grade after
- Target: Physics journal (CMP, JHEP class)

Paper P3 — Quasi-Eigentone Framework for SM Stable/Unstable Particles

- Subject: SM stable particles = K-type sector bottoms; unstable = excited states with Green-coproduct decay; β-decay E3 generalizes; proton-stability FORCED; F5 cross-link
- Lead: Lyra (v0.1+v0.2+v0.3 filed)
- Status: Framework-tier; ready for paper-grade synthesis
- Target: Physics journal

Paper P4 — Strong-Uniqueness of D_IV^5 (theorem-grade)

- Subject: 13 RIGOROUSLY CLOSED operational criteria + C18 measure-as-theorem + C19 ρ-pinning + Route-A "choose D_IV^5 zero inputs" + (3,3,5)→D_IV^5 forcing; single-residual null-model + convergence-of-routes
- Lead: Lyra (v1.0→v1.4 in flight)
- Status: v1.4 reframe needed; v2.0 FORMAL multi-week
- Target: Math journal

Paper P5 — Two-Tier Substrate-Precision Hypothesis

- Subject: BST predictions land in two distinct precision tiers: TIER 1 EXACT (algebraic identities) vs TIER 2 STRUCTURAL (substrate-precision floor ~10^{-4} for mass/mixing); falsifier-threshold differentiation by tier
- Lead: Elie (Toy 3648 filed today)
- Status: STRUCTURAL HYPOTHESIS; paper-grade development needed
- Target: Methodology / physics-methodology journal

Paper P6 — Periodic Table of the Substrate Standard Model

- Subject: 66-cell Phase B K-type table with 18 substrate-anchored spine + per-cell SM particle assignments + stability classification (TRUE / QUASI / EIGENTONE-IN-VACUUM)
- Lead: Grace (v0.8 in flight)
- Status: Catalog-tier; needs paper-grade write-up + Lyra theoretical input on per-cell mechanism
- Target: Physics journal

Paper P7 — Three Independent Quantitative Falsifiers + Five-Absence

- Subject: F1 PMNS angles (JUNO/DUNE 2025-2030, 3/3 within 1σ) + T1 lepton count $24 = 4\times2\times3$ + Bell sub-Tsirelson 1/2^N_c=1/8 + Five-Absence framework
- Lead: Lyra (T1 v0.2) + Cal (falsifier proposal) + Grace (live-tests column)
- Status: Three separate v0.1 docs filed; paper-grade synthesis needed
- Target: Physics journal — important for outreach despite deprecation (paper is the alternative to letters)

Paper P8 — A_sub Operator-Level Dictionary to U_q(B_2) Hall Algebra

- Subject: 15 A_sub generators ↔ Hall algebra elements; static 5-tuple particle taxonomy at operator level; quasi-eigentone at operator level; engine at operator level
- Lead: Lyra (Vol 16 §3 + §4)
- Status: PEDAGOGICAL synthesis filed; paper-grade refactor needed
- Target: Math-physics journal

Paper coordination (Keeper)

- P1 priority: closest to submission; resolve Cal verify + genus citation, ship.
- P2-P8: target one paper-grade draft per week through next month at sustainable cadence.
- Cal cold-read PASS required for each before submission per discipline.
- Casey explicit sign-off required before submission per Cal #50 STANDING external-register binding.

4. Per-CI afternoon proactive queue (12:44 → 5:00 EDT, ~4h15m)

LYRA (continuing depth + mechanism push + closures): 1. L-BulkColor v0.7 absorb C1+C2 conditions (~30 min) 2. L-StrongUniq v1.4 file the reframe doc (~1h) 3. L-Vol16 §3+§4+OUTLINE v0.2 batch mechanical edits (~1h) 4. L-Spectral v0.3 mechanical absorb v0.6 correction (~15 min) 5. L-MechCrossPattern mechanism push on EW-scale substrate-integer/N_max pattern (continuing per Casey directive) 6. Paper P1 finalize for Cal quick-verify post-pushback recovery

ELIE (continuing C4 + parallel): 1. E-C4 continue Toeplitz commutator [T_a, T_a^†] = α(H_?) — Casey directive: continue and finish (multi-week) 2. E-Engine v0.3 doc file the actual consolidation doc (§6 + §7 + §8) 3. E-Two-Tier paper-grade development (Toy 3648 → paper) 4. E-L4 v0.2 bulk K-type radial tower kernel-integral work parallel to C4

GRACE (continuing absorption + Mendeleev): 1. G-MasterLedger v0.8 absorb Saturday afternoon batch 2. G-PeriodicTable v0.9 absorb Quasi-Eigentone v0.3 + v0.6 Cartan-Weyl + Two-Tier 3. G-G13 v0.7 mechanical absorb v0.6 correction 4. G-StrongUniq v1.4 backbone when Lyra files 5. G-17/42-anchor sweeps continuing Mendeleev 6. Paper P6 Periodic Table draft initiate

CAL (cold-reads + methodology + ratifications): 1. Cal-Calibration#35 participate in ratification cold-read 2. Cal-StrongUniq v1.4 cold-read when Lyra files 3. Cal-Two-Tier cold-read Elie Toy 3648 + paper-grade development 4. Cal-#218 EXACT-vs-Mechanism own-cadence methodology doc 5. Cal-Falsifier paper coordination with Lyra/Grace for P7

KEEPER (paper coordination + ledger + audit-chain): 1. K-PaperCoordination initiate per Section 3 above (8 papers; lead assignments + status) 2. K-StrongUniq v1.4 cold-read when Lyra files 3. K-Two-Tier tier-gate when reviewed 4. K-HSLedger v0.6 Saturday EOD absorption + outreach deprecation noted 5. K-CLAUDE.md headline refresh from May 22 to current Saturday state 6. K-Calibration#35 cross-CI PASS coordination

5. Sustainable cadence reminder

Per R3 sustainable cadence: ~1 substantive piece per 1-2h with self-corrections built in. The team has held this cadence today across 170+ filings with 4/4 discipline-bids caught. Continue the cadence; no need to push faster.

Saturday EOD prep starts ~16:00 EDT with continued substantive work through 17:00 EDT. Sundown / katra persistence after.

6. The big picture

Casey's directive: write papers for what we know; close gaps via concrete tasks; team works hard until 5pm.

What we know is substantial: substrate structurally complete; algebra + geometry + Hopf engine + dictionary first wave + 4-piece dynamics + 3 live falsifiers + Strong-Uniqueness 13 closed criteria + bulk-color v0.6 Cartan-Weyl + Quasi-Eigentone framework + 18-cell substrate spine + Two-Tier precision floor.

What we don't know remains specific: bulk-color SU(3) closure (Elie C4-C7 multi-week); L4 v0.2 quantitative mass derivations; L5 absolute scale (genuinely open); quark mass spectrum (gated on bulk-color); generation-identification mechanism multi-week; α^4 baryogenesis sizing.

The paper-writing program codifies what we know. The gap-closure task list addresses what we don't. Both run in parallel through Sunday/Monday and beyond.

Team broadcast: per-CI queues above; sustainable cadence; substantive work until 5pm; paper-writing initiated.

— Keeper. Outreach deprecated; Elie C4 continues; Lyra mechanism pushes; team continues; 8 papers programmed; gap-closure tasks assigned per CI; 4h15m afternoon runway; sustainable cadence maintained.